

Scalability & Future Plan

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Team ID	SWTID-2026-8809
Project Name	EduGenie: Google Gemini Powered Learning Assistant
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Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local execution of CPU-heavy NLP model (LaMini-Flan-T5) leading to latency during simultaneous requests.	High resource usage on host device and potential delays for multiple concurrent users.	High
2	Single-session memory with no persistent database for user history or progress tracking.	Users cannot save past quizzes, summary notes, or customized learning path recommendations.	High
3	Input restricted to plain text via simple web interface; lack of multimodal document/image parsing.	Students cannot upload PDF textbooks, images, or study material directly for instant processing.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance FastAPI server running locally on Uvicorn.	Deploy containerized microservices on Kubernetes with auto-scaling to distribute API traffic.
2	Data Storage	In-memory processing without active persistent storage backend.	Integrate PostgreSQL for user profiles/ progress and Redis for prompt/response caching.
3	Performance	Local CPU inference for explanation module (LaMini model).	Migrate local models to dedicated GPU cloud instances or serverless inference endpoints.
4	Security	Basic client-server requests without user authentication or rate limits.	Implement OAuth2 / JWT authentication, API rate limiting, and encrypted environment secret storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Multimodal Support (PDF/Image OCR input) & persistent PostgreSQL user dashboard.	1 - 2 Months	Enables direct document summarization and saves student progress.
Phase 2	Voice-based interactive learning & Multilingual language translation.	3 - 4 Months	Improves accessibility for diverse and hands-free learners globally.
Phase 3	Mobile App Development (iOS/Android) with offline caching capability.	5 - 6 Months	Delivers seamless, anytime-anywhere learning experience across mobile devices.
Phase 4	Gamified Learning (Streaks, Badges) & LMS Integration (Google Classroom / Moodle).	7 - 8 Months	Boosts engagement and enables easy adoption in institutional workflows.